

The Complexity-Interpretability Tradeoff Between Risk-Scoring Methods: Implications for State Medicaid Agencies

Shailesh J. Divey¹, Nickolas K. Freeman², Dwight Lewis Jr.², Jason Parton²

¹The Ohio State University, ²The University of Alabama, Tuscaloosa

Problem definition: Healthcare systems increasingly rely on risk-scoring models to predict costs and guide resource allocation, particularly in large public programs such as Medicaid. However, the design of these models involves a fundamental tradeoff: more complex models often achieve higher predictive accuracy but are difficult to interpret, while simpler models are more transparent but may sacrifice performance. This complexity–interpretability tradeoff poses a central challenge for decision makers who must balance predictive accuracy with transparency, accountability, and operational usability. Despite widespread adoption of risk-scoring systems, there is limited understanding of how model complexity affects downstream decision quality in resource-constrained allocation settings. **Academic/practical relevance:** This study contributes to the intersection of operations management, machine learning, and healthcare analytics by explicitly linking predictive modeling choices to operational decision-making outcomes. While prior work evaluates models primarily based on predictive performance, we adopt a system-level perspective in which risk scores serve as inputs to resource allocation decisions. This approach aligns with emerging healthcare operations frameworks that emphasize interactions across stakeholders and the role of analytics in enabling efficient and equitable care delivery. Practically, healthcare administrators and policymakers must deploy models that are not only accurate but also interpretable enough to support adoption, satisfy regulatory requirements, and enable effective decision making. Understanding this tradeoff is therefore critical for designing implementable analytics systems in healthcare. **Methodology:** We develop a decision-analytic framework by using a large-scale longitudinal dataset covering more than 1.2 million Medicaid beneficiaries and over 12 million observations. We construct and compare a broad class of predictive models, including linear regression, decision trees, ensemble methods, and neural networks. A factorial experimental design systematically varies model features and inputs to isolate the impact of model complexity on predictive performance and interpretability. We then embed these predictive outputs into a resource allocation framework in which risk scores guide the allocation of preventive care interventions under budget constraints. This integration enables us to evaluate model performance not only in terms of predictive accuracy but also in terms of downstream decision quality and cost implications. **Managerial implications:** Developing risk scoring methods for healthcare involves a careful balance between model complexity and interpretability. While complex models might offer higher predictive accuracy, they need to be balanced with the need for transparent and explainable predictions that align with the clinical context and decision-making process. By combining the insights from these models with local knowledge and community engagement, resources can be allocated in a targeted and equitable manner, ultimately improving healthcare access and outcomes for underserved communities.

Key words: *risk scoring, predictive analytics, interpretability, fairness, healthcare resource allocation*
